INCH-POUND
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MILITARY SPECIFICATION

TANKS, TESTING, TIRE AND TUBE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of testing tank used for locating air leaks in inner tubes and mounted tubeless tires.

1.2 Classification.

1.2.1 Sizes. The tanks shall be of the following sizes, as specified (see 6.2.1):

Size 1 - Tank shall have minimum inside dimensions of 52 inches in diameter and 17.5 inches in depth
Size 2 - Tank shall have minimum inside dimensions of 72 inches in diameter and 30 inches in depth

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS), and supplement thereto, cited in the solicitation (see 6.2.1).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-EST-P, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4910

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SPECIFICATIONS

MILITARY

MIL-V-45554 - Vulcanizing Equipment, Including Related Items, Tire and Tube Rebuild and Repair, Preparation for Delivery of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods.
MIL-STD-1190 - Minimum Guidelines for Level C Preservation, Packing and Marking

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from DODSSP, Standardization Document Order Desk, 700 Robbins Avenue, Bldg 4D, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2.1).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E380 - Standard for Metric Practice.
ASTM D3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3 and 6.2.1).

3.2 Design. The testing tanks shall be new and of the manufacturer's latest design meeting the requirements specified herein. The testing tank shall be designed for detecting air leaks in inner tubes and mounted tubeless tires. Parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, and replacement without modification, distortion, or damage to the tanks or any tank part or component.

3.2.1 Measurement system. The US Customary System of Units (US) or the International System of Units (SI) may be used in the design and construction of the tanks. In this specification, all measurements, dimensions, sizes, and capacities are given in the US system. These measurements may be converted to the SI system by using the conversion factors and methods specified in ASTM E380.

3.2.2 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.3.1 Reclaimed materials. The manufacturer may use reclaimed materials for fabricating new parts. Reclaimed materials shall be reprocessed, remanufactured, or recycled in a manner that restores them to the same chemical composition and physical properties as the raw materials originally selected for use. Use of reclaimed parts as is or rebuilt from scrap or other used equipment shall not be permitted.

3.4 Construction. The testing tanks shall be furnished complete so that they can be used for the operations specified herein. Surfaces of all parts shall be clean and free of extraneous materials. External surfaces shall be smooth and free of sharp edges. Functional surfaces shall be free from burrs and defects affecting proper performance.

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3.4.1 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

3.4.2 Castings and forgings. Castings shall be free from visible blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from visible scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability.

3.4.3 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain all the performance requirements of the welded, brazed, or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.4.4 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

3.4.5 Corrosion resistance. Metallic components shall be inherently corrosion resistant, or shall be plated or treated to resist corrosion.

3.4.6 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

3.5 Performance and product characteristics.

3.5.1 Tank. The tank shall consist of a tank for containing liquid, with the capability for complete horizontal submergence and retention, in the submerged position, of inflated inner tubes and mounted tubeless tires within the size capacity specified in 3.5.2. The tank shall be designed to permit visual detection of air leaks in the inner tubes and mounted tubeless tires, while allowing the testing operation to be performed and controlled by one operator.

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3.5.2 Capacity. Capacity of the tanks shall be as specified below, without obstructing the view of the operator when observing for air leaks:

Size 1 - The testing tank shall have the capacity to completely submerge inflated inner tubes and mounted tubeless tires up to, and including, 12.00 X 24 size, with minimum inside dimensions of 52 inches in diameter and 17.5 inches in depth.

Size 2 - The testing tank shall be similar to size 1 except it shall have the capacity to completely submerge inflated inner tubes and mounted tubeless tires up to, and including, 18.00 X 24 sizxe, with minimum inside dimensions of 72 inches in diameter and 30 inches in depth.

3.6 Components. The tanks shall consist of, but not necessarily be limited to, the principle components specified in 3.6.1 through 3.6.5.

3.6.1. Tank, sizes 1 and 2. The tank shall be fabricated from galvanized or aluminized sheet steel, sheet aluminum or a nonmetallic material thick enough to preclude any deformation caused by the liquid and loads imposed during testing, and shall be capable of supporting the tube submerging mechanism. The tank shall be leak-proof when filled to the top with liquid and shall be reinforced throughout with corrosion resistant reinforcing. A drain plug or drain valve shall be provided to completely drain the tank while the tank is in an installed position.

3.6.2. Bridge. The bridge shall provide a support across the top of the tank to carry the air cylinder and spider specified in 3.6.3 and 3.6.5.

3.6.3. Air cylinder. The air cylinder unit shall be mounted in the bridge and shall operate, as specified in 3.5.1, within a line pressure ranging from 100 pounds per square inch gage (psig) to 175 psig. The cylinder shall be capable of withstanding without damage or leakage air line pressure of 200 psig for a period of two minutes. The cylinder may be single or double acting. The single acting cylinder shall be provided with two coil extension springs to act as retrievers, and shall be controlled by a two way air valve or an intake valve and an exhaust valve. The double acting cylinder shall be controlled by a three way air valve.

3.6.4. Air valve. The valve operating mechanism shall be mounted in an accessible position for hand operation without the use of tools and shall control the air flow in the cylinder. The air valve and cylinder shall be capable of holding the spider and tube at any required depth with a maximum allowable rise of 1/2 inch during a 20 minute period.

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3.6.5. Spider. The spider shall have the pattern of a hub with a minimum of six spokes and shall be attached to the air cylinder piston rod in a manner to permit the rotation of the submerged inner tube. Clearance (distance between the spider and the top of the tank) shall be large enough to permit unrestricted loading of the specified inflated inner tube sizes. The spider shall be designed so that it will not damage the tubes being tested.

3.6.6 Accessory support stand for size 1. When specified (see 6.2.1), the size 1, tank shall be provided with a support stand (see 3.3). The stand may be either a welded structural part of the tank or a separate holding (support) stand. The stand shall be constructed of steel incorporating not less than four legs to maintain the testing tank in a stable position during tube testing operations. The stand shall position the water tank at a height of 36 ±6 inches when measured from the water tank top edge to the floor level (surface on which the legs are sitting). The stand shall be capable of supporting the fully loaded tank without deformation or damage.

3.7 Product identification. The tanks shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

3.8 Workmanship. Standards of workmanship shall assure that the tanks shall have the stability, strength, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements, (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. When specified (see 6.2.1) first article inspection shall be applied to a preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1) first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract and shall consist of (a) through (e) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Performance test (see 4.6.1).
(c) Excessive pressure test (see 4.6.3).
(d) Tank leakage test (see 4.6.4).
(e) Packaging inspection (see 4.7).

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed as listed with all samples selected randomly.

Inspection or Test	Lot or Batch Size	Sample Size
Product examination (4.5)	1 - 15	5
Performance tests (4.6.1)	1 - 15	5
Excessive pressure test (4.6.3)	1 - 15	100%
Tank leakage test (4.6.4)	1 - 15	5
Packaging inspection (4.7)	1 - 15	3

The lots shall not exceed the maximum sizes indicated above. If lot size is less than or equal to sample size, 100% inspection is required. Each lot shall be accepted with no defects and rejected if one or more defects are found.

4.5 Product examination. Visually, dimensionally, and manually examine each tank to determine conformance with the requirements of 3.2 through 3.8. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Dimensional examination includes measuring dimensions as specified. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacturing at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operation. Failure of the contractor to provide objective evidence that the tank and its components have

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passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection. In addition, failure of the contractor to provide objective evidence that all parts are manufactured to definite standards, clearances, and tolerances so that no replacement part will degrade the form, fit, or function of the end item (see 3.4.4), shall be cause for rejection.

4.6 Tests.

4.6.1 Performance test. The tank shall be filled with liquid to the depth recommended by the manufacturer and the air inlet valve connected to a source capable of delivering compressed air at a pressure to comply with 3.6.3. An inner tube of appropriate size shall be inflated and placed in position under the spider. A tube testing operation shall be performed on the inner tube and the tank observed for compliance with the following:

(a) Functional requirements (3.5.1).
(b) Capacity (3.5.2).
(c) Air pressure (3.6.3).
(d) Air cylinder operation (3.6.3).
(e) Air valve operation (3.6.4).
(f) Depressor spider operation (3.6.5).

4.6.2 Air cylinder leakage test. A 12.00 X 24 inflated inner tube shall be submerged by the spider to the extent necessary to cover the highest projection of the tube with one inch of water (size 1). An 18.00 X 24 inflated inner tube shall be used for size 2. When the tube has been submerged to the specified depth, the cylinder air inlet valve shall be closed. The location of the spider, from the top of the tank, shall be measured. After a period of 20 minutes, the location of the spider shall be measured again at the same point. A variation of more than 1/2 inch in the location of the spider shall be cause for rejection of the tank (see 3.6.4).

4.6.3 Excessive pressure test. The tank air cylinder shall be connected to an air source of at least 200 psig. Blocks or any applicable retaining method shall be used against the air cylinder piston rod. The air valve shall be opened, permitting air pressure of at least 200 psig to be applied on the air cylinder piston for a period of at least two minutes. A soap and water solution shall be applied to the air cylinder connection and piston rod seal fitting. Any evidence of leakage or damage shall be cause for rejection (see 3.6.3).

4.6.4 Tank leakage test. The tank shall be filled with water to within one inch of the top, allowed to stand for a period of one hour, and then examined for evidence of leakage. Any evidence of leakage shall be considered cause for rejection of the tank (see 3.6.1).

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

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5. PACKAGING

5.1 Disassembly. The tanks may be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly without special skills or tools. Disassembly, when applicable, shall be as specified in MIL-STD-1186.

5.2 Preservation. Preservation shall be level A, C or Commercial, as specified (see 6.2.1).

5.2.1 Level A. Level A preservation shall be in accordance with requirements as specified in MIL-V-45554.

5.2.2 Level C. Level C preservation shall be in accordance with the requirements as specified in MIL-STD-1190.

5.2.3 Commercial. Commercial preservation shall be in accordance with the requirements as specified in ASTM D3951.

5.3 Packing. Packing shall be level A, B, C or Commercial, as specified (see 6.2.1).

5.3.1 Level A or B. Level A or B packing shall be in accordance with the requirements as specified in MIL-V-45554.

5.3.2 Level C. Level C packing shall be in accordance with the requirements as specified in MIL-STD-1190.

5.3.3 Commercial. Commercial packing shall be in accordance with the requirements as specified in ASTM D3951.

5.4 Marking. All marking, whether military or commercial, for interior and exterior containers shall be in accordance with the requirements as specified in MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The complete submersion type tanks are intended to be used to detect and locate air leaks in inner tubes, tubeless tires, valve stems, and valve covers in horizontal position.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).
(c) Size required (see 1.2.1).
(d) First article when required (see 3.1).
(e) Painting and finishing if different (see 3.4.6).
(f) Support stand when required (see 3.6.6).
(g) Product identification if different (see 3.7).
(h) First article inspection if different (see 4.3).
(i) Quality conformance inspection if different (see 4.4).
(j) Level of preservation, packing, and marking if different (see 5.2, 5.3 and 5.4).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

6.4 First article. When a first article is required, it shall be tested and approved under the appropriate provision of the Federal Acquisition Regulation. The first article should be a preproduction sample, or a standard production item from the contractor's current inventory. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for examination, test and approval of the first article.

6.5 Subject term (key word) listing.

Tank
Bridge
Spider
Air cylinder

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AL
Navy - YD
Air Force - 99

Preparing activity:
Army - AL

Project No. 4910-0739

Review activities:
Navy - MC
Air Force - 82
DLA - CS

User activity:
Army - ME

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